

Empirical indications from SNe Ia and H II galaxies toward a dynamical-speed-of-light cosmology

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ABSTRACT

Recently, Son et al. [MNRAS 544, 975 (2025)] proposed a progenitor-age-bias correction to standardized SNe Ia luminosities. Using the phenomenological flat w CDM, we find that the age-bias-corrected SNe Ia Hubble diagrams favor the $(w^* = -1/3, \Omega_m^* = 0)$ point as a fixed model by $\Delta\text{BIC} \simeq 13$. Remarkably, this point is also consistent with the H II galaxy constraints up to $z \sim 14$ recently reported by Chávez et al.. The $(w^* = -1/3, \Omega_m^* = 0)$ point admits a compact *zero-parameter* luminosity-distance relation $d_L = \frac{c}{H_0} (1+z) \ln(1+z)$, once H_0 is calibrated using local probes. The $\ln(1+z)$ factor can by itself account for the observed excess moduli in high-redshift SNe Ia, without requiring dark energy. More broadly, beyond flat w CDM, the same $\ln(1+z)$ factor arises naturally along the degenerate branch of Dolgov–Barrow cosmologies, characterized by power-law expansion $a = (t/t_0)^{\frac{2}{3(1+w)}}$ and *dynamical speed of light* $c = c_0 a^{-\frac{1+3w}{2}}$, which possess a *non-trivial kinematic relation* $c = \frac{c_0}{H_0} \dot{a}$ between the speed of light and the cosmic expansion rate. Accordingly, these recent indications from age-biased-corrected SNe Ia and high- z H II galaxies may call for a structural shift beyond the Λ CDM paradigm. In particular, the 1998 discovery of cosmic acceleration need not directly imply dark energy, but may instead point toward Dolgov–Barrow cosmologies, where the speed of light, rather than dark energy, becomes dynamical. As an alternative to $w_0 w_a$ CDM, we put forth a dynamical-speed-of-light framework with concrete phenomenological predictions.

Key words. cosmology: observations – distance scale – supernovae: general – cosmological parameters

1. Motivation

Type Ia supernovae (SNe Ia) provided the first evidence for the accelerated expansion of the Universe and serves as a cornerstone of the dark-energy cosmological paradigm. Recently, Son et al. (2025) (hereafter, the YONSEI group) proposed a progenitor-age-bias correction (ABC) to standardized SNe luminosities. When considered within the phenomenological dynamical-dark-energy $w_0 w_a$ CDM model choice, the ABC brings the SNe into closer agreement with DESI BAO and CMB observations (Son et al. 2025). The YONSEI group’s work has ignited lively debates regarding the validity of the ABC and its cosmological consequences within the standard Λ CDM framework (Wiseman et al. 2026; Chung et al. 2026; Zhang et al. 2026; Sah et al. 2026a; Popovic & Sullivan 2026; Mirpoorian et al. 2026; Mahtessian et al. 2026).

The YONSEI group’s ABC—if *valid*—could signal a structural rethinking in late-time cosmology beyond merely a re-evaluation of Λ CDM parameters, however. In Nguyen (2026a), we investigated the Pantheon+ and DES-SN5YR Hubble diagrams using the ABC. We conducted three complementary analyses showing that the post-ABC SNe data sharply converge to the “Kolb” $(w^*, \Omega_m^*) = (-1/3, 0)$ point in the w CDM parameter space, or equivalently complying to a compact distance relation

$$d_L = \frac{c}{H_0} (1+z) \ln(1+z). \quad (1)$$

The $\Delta\text{BIC} \simeq 25$ over the Λ CDM. This finding, conditioned on the validity of the ABC, has three concrete consequences:

First, based on the logarithmic relation, we made a *falsifiable prediction* for SNe within $1 \lesssim z \lesssim 2$, a redshift range already accessible within current capabilities. Upon reviewing the recent literature, we find that a potential avenue for falsifying this prediction already exists: the H II-galaxy Hubble diagram

of Chávez et al. (2026), which extends to $z \sim 14$ and is independent of the YONSEI’s SNe ABC (see also Terlevich et al. (2015); González-Morán et al. (2019); Chávez et al. (2025)). We shall therefore confront these H II galaxy constraints with our prediction, and further suggest a direct test of the logarithmic relation using H II galaxies data.

Second, Son et al. (2025) employed the Chevallier–Polarski–Linder (CPL) parametrization to analyze the ABC-corrected SNe together with DESI BAO and CMB observations. Such an extension, however, may not be necessary for the SNe themselves: as shown in Nguyen (2026a), the compact logarithmic relation (1) already constitutes a highly efficient parsimonious *zero-parameter* description, once H_0 is calibrated using local probes. This Kolb point has a privileged status that has been overlooked within the flat w CDM model itself. We shall therefore revisit the Kolb point in the flat w CDM parameter space.

Third, the CPL parameterization introduces dynamical dark energy through w_a . The logarithmic relation suggests a qualitatively different avenue. The Kolb point is degenerate... The same relation naturally arises from a distinguished class of Dolgov–Barrow cosmologies, in which the speed of light is dynamical on the expanding cosmic background. We shall therefore expose this connection and, as an alternative to w_a and dynamical dark energy, introduce a cosmological framework where the speed of light becomes dynamical.

Our Letter is structured as follows. Sect. 2 revisits and expands our w CDM analysis of the SNe Hubble diagrams. Sect. 3 discusses the privileged Kolb point, while Sect. 4 compares it with the H II-galaxy constraints of Chávez et al. (2026). Sect. 5 turns to the Dolgov–Barrow cosmologies and their connection with the logarithmic relation. Sect. 6 proposes a connection between w CDM and the Dolgov–Barrow cosmologies. Sect. 7 clarifies the role of BAO and CMB probes, while Sect. 8 expands the physical consequences of our proposed framework.

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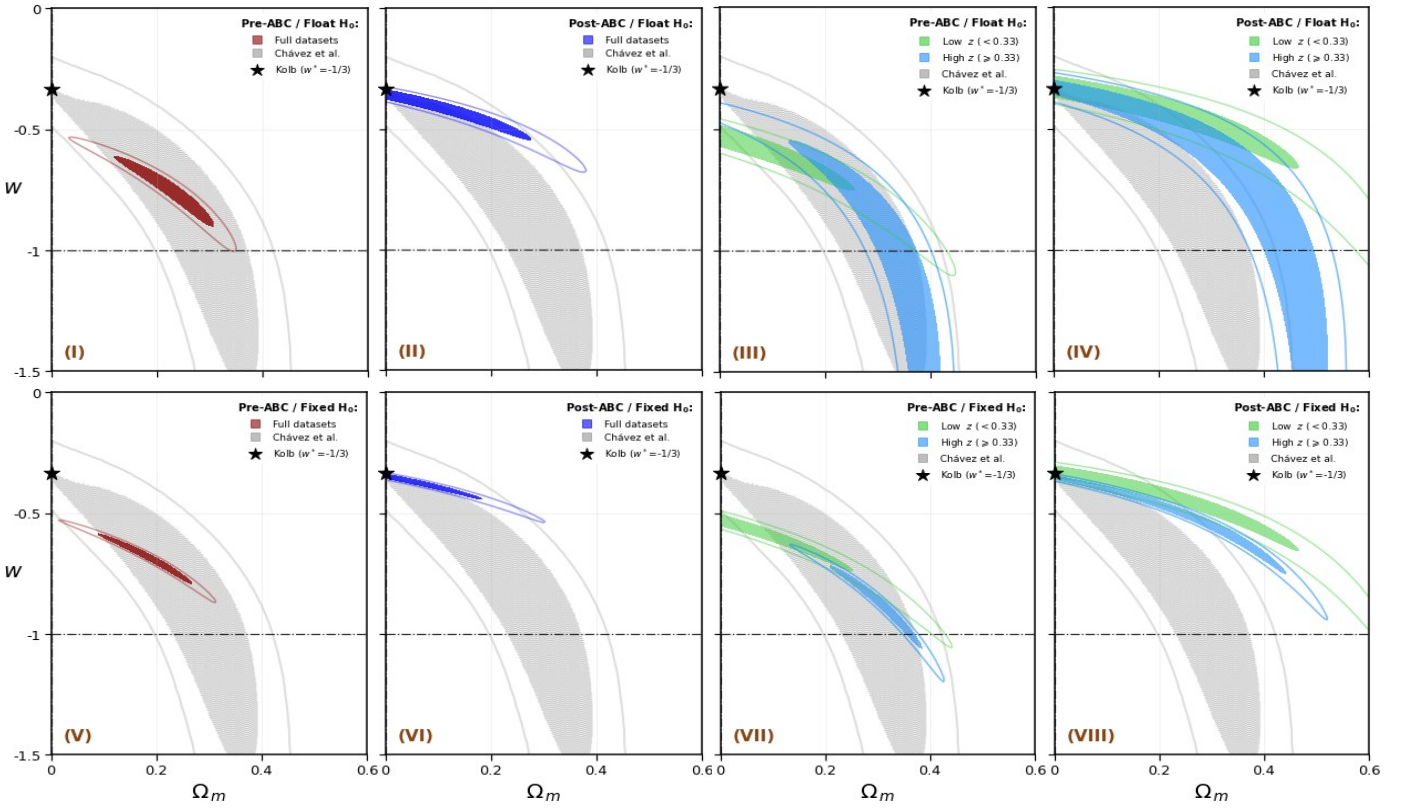


Fig. 1: (See Section 4 for description of Chávez et al.’s confidence regions.)

2. Revisiting the post-ABC SNe Hubble diagram using flat w CDM

In one of our analyses in [Nguyen \(2026a\)](#), we handled the post-ABC Pantheon+ and DES-SN5YR Hubble diagrams within flat w CDM. Here, we shall revisit that analysis in a more comprehensive view. Since the two SNe compilations possess independent absolute calibrations, we construct the joint likelihood

$$\chi^2 = \chi^2_{\text{Pan}+} + \chi^2_{\text{DES}}, \quad (2)$$

allowing an independent H_0 normalization for each dataset. Details of the likelihood construction and covariance treatment are given in Appendix B of [Nguyen \(2026a\)](#).

The resulting confidence regions in the (w, Ω_m) plane are shown in Fig. 1. Panels 1 and 2 reproduce paper 1, with an addition of the Chávez contours (sect 4). We redraw using OM instead of ODE and restrict $\Omega_m \geq 0$. Panel (I) displays the original SNe data missing the Kolb point, while Panel (II) shows the result after applying the ABC. The post-ABC confidence regions shift markedly toward the Kolb point $(w^*, \Omega_m^*) = (-1/3, 0)$, with the best-fitting w CDM model lying in its immediate vicinity, $(\hat{w}, \hat{\Omega}_m) = (-0.359, 0.006)$; see the fourth line in Table 1. When

treated as a fixed model, the Kolb point is favored over the best-fitting w CDM model by $\Delta\text{BIC} \approx 13 - 15$, whether the analysis is performed on the full sample or on the low- z and high- z subsamples; see Table 1.

To examine whether this convergence is driven primarily by either the low- or high-redshift SNe, we divide the combined sample at $z_{\text{split}} = 0.33$, yielding two subsets of approximately equal size ($N_{\text{low}} = 1754$ and $N_{\text{high}} = 1767$). Their confidence regions are shown in Panels (III) and (IV) of Fig. 1. Both redshift subsets independently remain directed toward the vicinity of the Kolb point, indicating that the convergence is not produced solely by one particular redshift range.

Since H_0 in principle can be calibrated using local probes, we repeat the analysis after fixing the absolute normalization using only the low-redshift SNe. The resulting confidence regions are displayed in the lower four panels of Fig. 1. In particular, Panels (VI) and (VIII) show an even sharper alignment with the Kolb point, demonstrating that its preferred location is not an artifact of freely adjusting the H_0 normalization of the Hubble diagram.

For reference, Fig. 1 also shows as a backdrop the independent H II-galaxy confidence regions digitized from the panel (w_0, Ω_m) in Fig. 6 of [Chávez et al. \(2026\)](#); these will be discussed

 Table 1: Comparison between flat w CDM best fit and the Kolb point for the Pantheon+ and DES joint sample.

	Range of z	$N^{\text{Pan}+}$	N^{DES}	N^{tot}	$\hat{H}_0^{\text{Pan}+}$	$\hat{H}_0^{\text{DES}}$	$\hat{w}$	$\hat{\Omega}_m$	$\hat{H}_{0(\text{Kolb})}^{\text{Pan}+}$	$\hat{H}_{0(\text{Kolb})}^{\text{DES}}$	$\chi^2_{w\text{CDM}}$	χ^2_{Kolb}	$\Delta\chi^2$	AIC $_{w\text{CDM}}$	AIC $_{\text{Kolb}}$	ΔAIC	BIC $_{w\text{CDM}}$	BIC $_{\text{Kolb}}$	ΔBIC
Pre-ABC	$0 < z < 2.3$	1701	1820	3521	72.70	69.13	-0.750	0.226	70.64	65.66	3380.2	3583.6	+203.4	3388.2	3587.6	+199.4	3412.9	3599.9	+187.1
Pre-ABC	$0 < z < 0.33$	1283	471	1754	72.43	69.00	-0.528	0.000	71.14	67.38	1854.7	1901.7	+46.9	1862.7	1905.7	+42.9	1884.6	1916.6	+32.0
Pre-ABC	$0.33 \leq z < 2.3$	418	1349	1767	73.97	70.32	-1.124	0.356	68.49	64.73	1507.4	1525.1	+17.7	1515.4	1529.1	+13.7	1537.3	1540.0	+2.7
Post-ABC	$0 < z < 2.3$	1701	1820	3521	72.78	69.20	-0.359	0.006	72.56	68.79	3381.7	3384.9	+3.2	3389.7	3388.9	-0.8	3414.4	3401.3	-13.1
Post-ABC	$0 < z < 0.33$	1283	471	1754	72.53	69.10	-0.327	0.000	72.57	69.15	1857.8	1857.9	+0.1	1865.8	1861.9	-3.9	1887.7	1872.8	-14.9
Post-ABC	$0.33 \leq z < 2.3$	418	1349	1767	74.02	70.37	-0.819	0.405	72.24	68.59	1507.4	1508.6	+1.2	1515.4	1512.6	-2.8	1537.3	1523.5	-13.8

separately in Sect. 4. Taken together, the full-sample, redshift-split, and fixed-normalization confidence contours consistently identify the vicinity of $(w^*, \Omega_m^*) = (-1/3, 0)$ as a distinguished region of the post-ABC SNe Hubble diagram. In the following section, we show that this point possesses a special analytic property: its luminosity-distance relation is exactly logarithmic.

Table 1 for statistics and parameter estimates. Since the fixed- H_0 contours depend on the particular fiducial values adopted for the absolute normalization, we use them only as a visualization diagnostic; all quantitative model comparisons reported below are based on the floating- H_0 analysis.

3. The privileged Kolb point ($w^* = -1/3, \Omega_m^* = 0$): A logarithmic distance relation and $q = 0$

In flat w CDM ($\Omega_M + \Omega_{DE} = 1$), the luminosity distance is

$$d_L = \frac{c}{H_0} (1+z) \int_0^z \frac{du}{\sqrt{\Omega_m(1+u)^3 + \Omega_{DE}(1+u)^{3(1+w)}}}. \quad (3)$$

Importantly, every member on the line $\Omega_m = 0$ represents a single-fluid cosmology which admits the following power-law expansion and distance relation

$$a(t) = a_0 (t/t_0)^{\frac{2}{3(1+w)}}; \quad d_L = \frac{2c(1+z)}{H_0(1+3w)} \left[1 - (1+z)^{\frac{1+3w}{2}} \right]. \quad (4)$$

where $t_0 \equiv \frac{2}{3(1+w)H_0}$, the current age of the Universe.

Precisely at the point $(w^* = -1/3, \Omega_m^*) = (-1/3, 0)$, Eq. (4) recovers Eq. (1). We attribute this special point to Kolb, who first proposed a form of K-matter with an equation of state (EoS) $w = -1/3$ (Kolb 1989). The resultant universe expands linearly, $a \propto t$, to which Kolb referred as “a coasting universe”.

Thus the logarithmic relation is not an external ansatz imposed on flat w CDM: it is the exact luminosity-distance relation at its single-fluid $w^* = -1/3$.

Its location is geometrically distinguished. For generic flat w CDM, the deceleration parameter is

$$q \equiv -\frac{a\ddot{a}}{\dot{a}^2} = -\frac{1}{2} \left(1 + \frac{d \ln H}{d \ln a} \right) = \frac{1}{2} + \frac{3w}{2} \frac{\Omega_{DE}}{\Omega_m a^{3w} + \Omega_{DE}} \quad (5)$$

Only on the single-fluid boundary $\Omega_m = 0$ is q constant throughout the evolution, with $q = \frac{1}{2}(1+3w)$. Within flat w CDM, $(w^* = -1/3, \Omega_m^* = 0)$ is therefore privileged both as the transition between decelerating and accelerating single-fluid expansion and as the unique point at which the luminosity-distance relation acquires a parsimonious logarithmic form. This result will be relevant to our Conclusion...

4. Chávez et al.’s H II-galaxy Hubble diagram up to $z \sim 14$ as a *retrodiction* of the Kolb point

Over the past decade, H II-galaxy studies have progressively extended the maximum redshift of the Hubble diagram, from $z_{\max} \simeq 2.33$ (Terlevich et al. 2015; González-Morán et al. 2019), to $z_{\max} \simeq 7.5$ (Chávez et al. 2025), and most recently to $z_{\max} \simeq 14$ (Chávez et al. 2026). Among their reported results, these analyses constructed joint likelihoods in the (w, Ω_m) plane within flat w CDM, after marginalizing over the relevant nuisance parameters.

Independently, in Nguyen (2026a) we predicted from the logarithmic luminosity-distance relation that the distance kernel

$$K(z) \equiv \frac{H_0}{c} \frac{d[d_L/(1+z)]}{d \ln(1+z)} = 1. \quad (6)$$

This prediction becomes particularly discriminating at high redshift, where the logarithmic relation departs increasingly from the corresponding Λ CDM kernel. Larger samples of SNe Ia in the range $1 \lesssim z \lesssim 2$ where the logarithmic relation deviates markedly from the kernel of Λ CDM. The H II-galaxy Hubble diagram therefore provides an independent high- z *retrodiction* of the relation selected by the post-ABC SNe.

Remarkably, the independent H II-galaxy confidence regions of Chávez et al. likewise extend toward the Kolb point. As shown in Fig. 1, their progressively higher-redshift constraints approach the region selected independently by the post-ABC SNe. If restricted to the single-fluid boundary $\Omega_m = 0$, the H II-galaxy likelihood lies in the vicinity of $w = -1/3$, with the Kolb point comfortably enclosed within the reported confidence region. Since the logarithmic relation was inferred from the SNe independently of these H II data, this agreement constitutes a *retrodiction* rather than an input to the construction. A rough visual estimate from the panel (w_0, Ω_m) in Fig. 6 of Chávez et al. (2026) indicates that, when restricted to the single-fluid line $\Omega_m = 0$, the approximate 2σ confidence range for w is

$$-0.50 \lesssim w \lesssim 0.20, \quad (7)$$

comfortably enclosing the Kolb value $w = -1/3$.

Further support comes from the recent analysis of Wei & Melia (2025), who fitted the high-redshift H II-galaxy Hubble diagram up to $z \sim 7.5$ directly with both a logarithmic luminosity distance and flat w CDM. They obtained $\chi^2 = 848.15$ for the logarithmic relation and $\chi^2 = 846.72$ for w CDM, a difference of only $\Delta\chi^2 = 1.43$. Thus, the very high redshift H II data hold promise to be statistically compatible with the Kolb point.

A particularly useful future test would be to determine whether this behavior is stable across redshift. For example, the H II-galaxy sample could be divided into low- and high- z subsets and fitted with the quadratic extension

$$\frac{d_L}{1+z} = \frac{c}{H_0} \ln(1+z) [1 + \delta \ln(1+z)], \quad (8)$$

for which the logarithmic relation corresponds to $\delta = 0$. Such an analysis, performed together with the appropriate H II-galaxy nuisance parameters to be treated properly by Chavez, would provide a direct test of both the Kolb point and its stability with redshift.

5. The dynamical- c Dolgov–Barrow cosmology

The logarithmic relation is not exclusive to Kolb. Rather, we shall show that it belongs to a broader class of FLRW cosmologies that involve a dynamical speed of light (DSL).

Consider a FLRW metric with dynamical speed of light c_{dyn}

$$ds^2 = -c_{\text{dyn}}^2 dt^2 + a^2(t) \left[\frac{dr^2}{1 - \kappa r^2} + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2) \right], \quad (9)$$

where $\kappa \in \{-1, 0, 1\}$ denotes spatially open, flat, closed universe respectively. Remarkably, the Ricci scalar curvature is expressible in a neat form

$$\mathcal{R} = \frac{6}{a^2} \left[\kappa + \left(\frac{aH}{c_{\text{dyn}}} \right)^2 \left(1 + \frac{d}{d \ln a} \ln \frac{aH}{c_{\text{dyn}}} \right) \right] \quad (10)$$

Next, consider a Barrow-type dynamical c (Barrow 1999)

$$c_{\text{dyn}} = c_0 (a/a_0)^{-\zeta}, \quad (11)$$

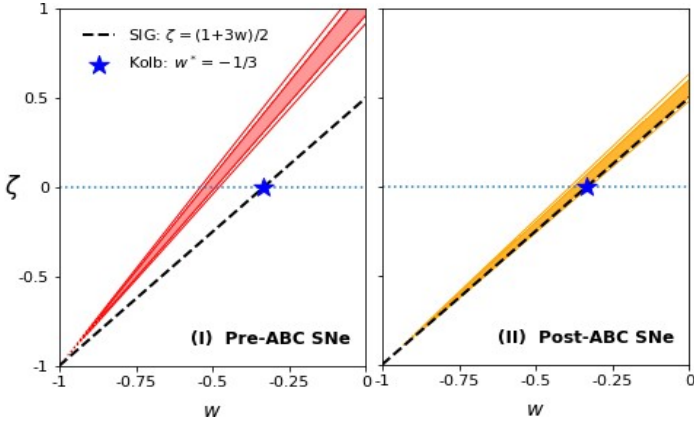


Fig. 2: Test

and a Dolgov-type power-law expansion (Dolgov 1997)

$$a(t) = a_0 (t/t_0)^{\frac{2}{3(1+w)}}. \quad (12)$$

The $a(t)$ function yields $H(a) \equiv \frac{\dot{a}}{a} = \frac{2}{3(1+w)t_0} (a/a_0)^{-3(1+w)/2}$ and the deceleration parameter $q \equiv -\frac{a\ddot{a}}{\dot{a}^2} = \frac{1+3w}{2}$. The Ricci scalar simplifies to

$$\mathcal{R} = \frac{6}{a^2} \left[\kappa + \left(\frac{a_0 H_0}{c_0} \right)^2 (1 + \zeta - q) \left(\frac{a}{a_0} \right)^{2(\zeta - q)} \right] \quad (13)$$

Remarkably, the line

$$\zeta^* \equiv q = \frac{1+3w}{2}, \quad (14)$$

results in a Ricci scalar proportional to a^{-2} . We therefore refer to this line as the scale-invariant geometry (SIG) line.¹

The Dolgov–Barrow cosmology given by Eqs. (11) and (12) admits the following luminosity-distance relation

$$d_L = \frac{c_0 (1+z)}{H_0} \frac{(1+z)^{\frac{\zeta - (1+3w)/2}{1+\zeta}} - 1}{\zeta - (1+3w)/2}; \quad t_0 \equiv \frac{2}{3(1+w)H_0} \quad (15)$$

The SIG line (14) corresponds to a logarithmic relation

$$d_L = c_0 t_0 (1+z) \ln(1+z) \quad (16)$$

Thus the same logarithmic distance relation selected by the post-ABC SNe arises along a one-dimensional branch of the Dolgov–Barrow family, hence a *degeneracy*.

Fig. 2 shows the corresponding SNe likelihoods in the Dolgov–Barrow (w, ζ) plane, before and after applying the ABC. The dashed line denotes the SIG condition $\zeta = (1+3w)/2$, along which the luminosity distance is exactly logarithmic. Prior to the ABC, the preferred region is displaced from this branch. After

¹ The origin of this special status for the SIG line can be made more transparent. In the Dolgov–Barrow cosmology (11) and (12), the radial null-geodesic obeys $dr = \frac{c dt}{a} \propto a^{-(1+\zeta)} dt \propto t^{-\frac{2(1+\zeta)}{3(1+w)}} dt$. Under a global dilation of cosmic time $t \rightarrow \lambda t$, then $dr \rightarrow \lambda^{1-\frac{2(1+\zeta)}{3(1+w)}} dr$. The radial null-geodesic is invariant *if and only if* $\frac{2(1+\zeta)}{3(1+w)} = 1 \iff \zeta = \frac{1+3w}{2}$, which is precisely the SIG condition (14), marking the unique *time-dilation-invariant branch*. Furthermore, along the SIG line, the comoving distance between the two cosmic events t_1 and t_2 , $r_{12} = \int_{t_1}^{t_2} \frac{c dt}{a} \propto \int_{t_1}^{t_2} \frac{dt}{t} \propto \ln \frac{t_2}{t_1}$, depends only on the ratio t_2/t_1 and is invariant under a global rescaling of both time instants. This multiplicative scale invariance is the root cause of the logarithm form (16) manifest along the SIG line.

the ABC, however, the SNe likelihood moves markedly toward the SIG line and becomes elongated along it, reflecting the logarithmic degeneracy derived above.²

This behavior is central to the construction of Sect. 6. In the flat w CDM plane, the post-ABC SNe likelihood approaches the Kolb point $(w, \Omega_m) = (-1/3, 0)$; in the Dolgov–Barrow plane, the same data approach the SIG branch. Since the Kolb point lies at the common intersection of these two descriptions, $(w, \Omega_m, \zeta) = (-1/3, 0, 0)$, the post-ABC likelihoods on the two planes can be joined continuously there. This provides the empirical basis for the ζ - w CDM construction introduced in the following section.

6. A transition from w CDM to Dolgov–Barrow cosmology: Dynamical dark energy or dynamical speed of light?

In the face of DESI and several ongoing cosmological tensions, a popular theme has been to promote dark energy into a dynamical entity (cite a review?). Among many possible choices, a default model is the CPL parameterization (Chevallier & Polarski 2001; Linder 2003) which augments the dark-energy sector with an *ad hoc* evolving equation of state, $w(a) = w_0 + w_a(1 - a)$.

The logarithmic relation (1), realized *both* at the Kolb point $(w^*, \Omega_m^*) = (-1/3, 0)$ within flat w CDM (Sect. 3) and along the SIG line within the Dolgov–Barrow cosmology (Sect. 5), offers a *structural shift* from the common dynamical-dark-energy theme: it invites a natural transition from w CDM into a new cosmological framework where the speed of light becomes dynamical.

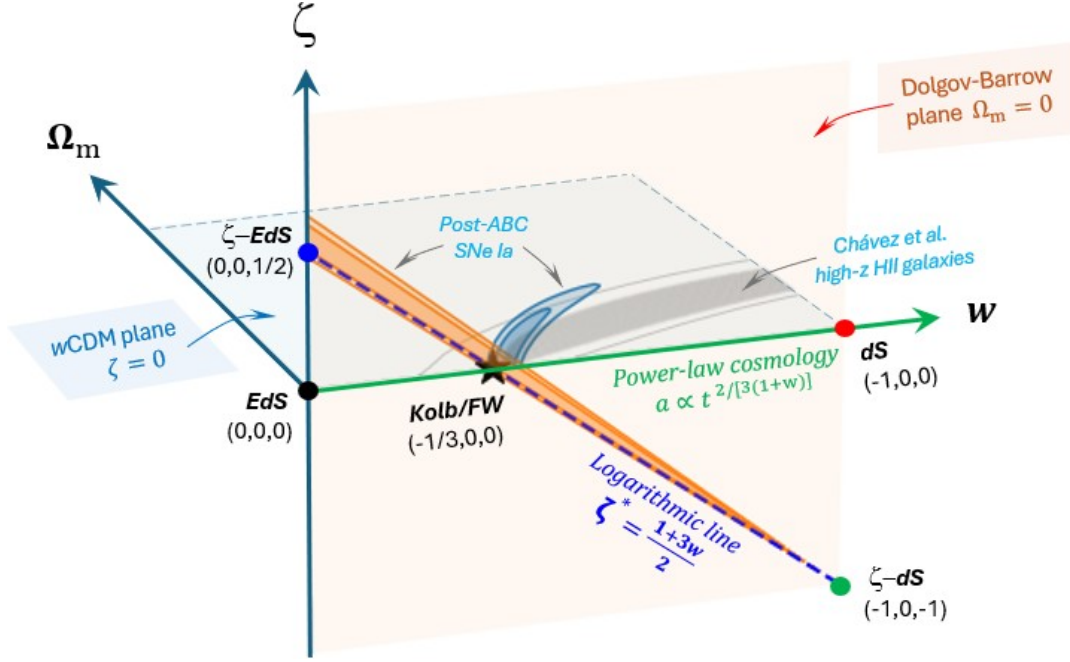
The two ingredients essential for this transition are:

(I) Interestingly, each member of the single-fluid line in flat w CDM corresponds one-to-one with a Dolgov-type power-law expansion $a \propto t^{\frac{2}{3(1+w)}}$. This aspect is important for our exposition in Sect. 6. The Dolgov–Barrow family seamlessly extends the single-fluid boundary of w CDM, viz. the $\Omega_m = 0$ line, into the dynamical-speed-of-light (DSL) direction, enabling a novel degree of freedom absent in the standard Λ CDM paradigm.

(II) Within the Dolgov–Barrow family, the SIG line $\zeta^* = (1+3w)/2$ singles out a distinguished scale-invariant geometry branch, which connects the empirically established logarithmic relation to an underlying scale-invariant gravitational theory. Importantly, the SIG line possesses a specific quantitative relation—and thus a *falsifiable* prediction—between the dynamical speed of light and the cosmic expansion rate, specifically $c_{\text{dyn}} = \frac{c_0}{a_0 H_0} \dot{a}$, as shown in Eq. (20).

The possibility of a *variable* speed of light (VSL) for *gravity* has a long history, originating with Einstein (1911) during his quest for GR, and revived in modern cosmological contexts (Moffat 1993; Albrecht & Magueijo 1999; Barrow 1999), notably to resolve the horizon paradox without invoking inflation; see the comprehensive VSL review by Magueijo (2003). The *dynamical* speed of light (DSL)—the central theme of this Letter—belongs to this general VSL umbrella. However, there is a fundamental difference: instead of being prescribed as an external function $c(x^\mu)$ of spacetime (or merely $c(t)$ as is customarily treated in the VSL literature), *our dynamical c is inherently linked to the scalar curvature of spacetime itself*. The dynamics of c hence stems directly from the dynamics of space-

² Notably, the HII-galaxy data are independent of YONSEI’s ABC. We therefore suggest that they be analyzed within the Dolgov–Barrow cosmology as an independent test of the SIG branch.


 Fig. 3: The triplet are (w, Ω_m, ζ) .

time's scalar curvature and is governed by a well-defined action of gravity and matter (Nguyen 2025c).³

Figure 3 summarizes our proposed structure, which we shall refer to as the ζ - w CDM framework. It inherits the flat w CDM and the Dolgov–Barrow parameter spaces, involving three cosmological parameters (w, Ω_m, ζ) . It joins these two parameter spaces along the single-fluid boundary $\Omega_m = 0$, which coincides with the power-law Dolgov line $a(t) \propto t^{\frac{2}{3(1+w)}}$. The horizontal w CDM (w, Ω_m) plane is a replot of Panel (II) of Fig. 1, showing the 1- and 2-sigma confidence regions of YONSEI's post-ABC SNe Hubble diagram and of Chavez et al.'s $z \sim 14$ H II galaxy constraints. Both confidence regions converge to the Kolb point $(w^*, \Omega_m^*, \zeta) = (-1/3, 0, 0)$. In the w CDM plane, the speed of light is a constant.

The vertical Dolgov–Barrow (w, ζ) plane is a replot of the right panel of Fig. 2, showing the 1- and 2-sigma confidence regions of post-ABC SNe Hubble diagram. The confidence regions of post-ABC SNe for the two planes join seamlessly at the Kolb point, with the independent H II-galaxy constraints of Chávez et al. lending additional support. The power-law cosmology line also contains the Einstein–de Sitter (EdS) universe⁴ $(w, \Omega_m, \zeta) = (0, 0, 0)$ where $a \propto t^{2/3}$; and the de Sitter (dS) universe $(w, \Omega_m, \zeta) = (-1, 0, 0)$ where $a \propto e^{H_0 t}$.

Importantly, in the Dolgov–Barrow plane, as proved in Sect. 5, there exists a distinguished SIG branch $\zeta^* = (1 + 3w)/2$ (dashed line), along which the luminosity distance is exactly logarithmic *regardless of w* , per Eq. (16). As the SIG line is degenerate with respect to the SNe and H II galaxy probes, ζ acts as a *tunable DSL index*, with its value to be determined by other cosmological probes, including BAO and the CMB. We should

stress that while the structure presented in Fig. 3 is phenomenological, the SIG branch can be generated from a scale-invariant action where matter couples with gravity via a dilatonic sector of spacetime Nguyen (2025b,c). A discussion of this particular DSL gravitational theory lies beyond the scope of this Letter.

Along the SIG branch, three anchors (ζ -EdS, Fujii–Wetterich (FW), and ζ -dS) stand out:

$$\begin{aligned} \zeta\text{-EdS} : (w, \zeta) &= (0, 1/2) \implies a \propto t^{2/3}, \quad c \propto a^{-1/2}, \\ \text{FW} : (w, \zeta) &= (-1/3, 0) \implies a \propto t, \quad c \equiv \text{const}, \\ \zeta\text{-dS} : (w, \zeta) &= (-1, -1) \implies a \propto e^{H_0 t}, \quad c \propto a. \end{aligned} \quad (17)$$

The ζ -EdS point is particularly interesting. It corresponds to an EdS expansion mode $a \propto t^{2/3}$ and a declining speed of light $c \propto a^{-1/2}$. As demonstrated in Nguyen (2025a), the 1998 discovery of late-time acceleration (Riess et al. 1998; Perlmutter et al. 1999), conventionally interpreted as evidence for dark energy, can arise from an EdS universe accompanied by declining c .

The FW point, closely related to the gravitational theory of Fujii (1982) and Wetterich (1988), deserves remark. Although the FW point shares the same kinematics as the Kolb point, viz. $a \propto t$ and $c \equiv \text{const}$, the FW and Kolb points obey fundamentally different gravitational dynamics. While the Kolb universe satisfies GR, viz. the Friedmann equations, with a new form of matter—the K-matter with an EOS $w = -1/3$ —driving the cosmic expansion, the FW universe operates in a scale-invariant theory of gravity and matter. Thus, the Kolb/FW point constitutes a common kinematic junction between the two frameworks, without requiring dynamical equivalence.

In summary, the convergence of the independent SNe and H II galaxy constraints toward the common Kolb/FW junction motivates exploring the dynamical- c direction as an alternative to dynamical dark energy. Since BAO and CMB observables depend on both the expansion history (encoded in w) and the propagation of light (encoded in ζ), these probes can, in principle, break the degeneracy along the SIG branch and determine ζ . SNe and H II galaxies single out the logarithmic branch, while BAO and the CMB select where along that branch the DSL index actually resides.

³ Note: our DSL approach closely reflects Einstein (1911)'s original vision of a VSL during his quest for GR. Therein, instead of making c a function of spacetime, he deliberately experimented it as a function of the dynamical gravitational potential Φ , via $c = c_0 (1 + \Phi/c^2)$. Whereas his version is not generally covariant, replacing the gravitational potential with spacetime's curvature scalar would restore general covariance.

⁴ The lines $w = 0$ and $\Omega_m = 1$ also correspond to EdS universe. A technical procedure to remove the EdS degeneracy will be reported elsewhere.

7. The role of BAO and CMB in determining ζ

The vanilla w CDM framework does not provide sufficient freedom to reconcile the post-ABC SNe constraints with BAO and CMB observations, as is a common issue with the DESI-era cosmological tensions. For example, in Son et al. (2025), the confidence regions from the SNe, BAO and CMB probes are in tension within w CDM. The w_a parameter of the CPL model was then used to bring these probes into closer agreement.

The ζ - w CDM framework developed here opens an alternative. Rather than invoking the w_a parameter of the CPL model, the tunable DSL index ζ enables an additional direction in parameter space. Standard w CDM is recovered on the $\zeta = 0$ plane, whereas allowing $\zeta \neq 0$ can, in principle, bring the SNe, BAO and CMB probes into agreement.

Viewed within the enlarged ζ - w CDM parameter space, the current tension between post-ABC SNe with BAO/CMB is an artefact of analyzing the observational data within the restricted $\zeta = 0$ plane. It is therefore plausible that *the three probes may converge away from this plane, at a non-zero value of ζ* . For example, the convergence could occur around the ζ -EdS point $(w, \Omega_m, \zeta) = (0, 0, 1/2)$, where the phenomenology of late-time acceleration is reproduced by a declining speed of light $c \propto a^{-1/2}$ on an expanding EdS background $a \propto t^{2/3}$, a result obtained in Nguyen (2025a).

Furthermore, as Sect. 6 demonstrated, SNe and H II galaxies cannot determine ζ along the *degenerate* SIG branch: their luminosity distance has the identical logarithmic form for every pair (w, ζ) satisfying $\zeta = (1 + 3w)/2$. BAO and the CMB hence step in to break this degeneracy. Since BAO and CMB observables depend on both the expansion history, controlled here by w , and the propagation of light, controlled by ζ , the BAO/CMB probes can, in principle, lift the SIG degeneracy and determine ζ .

A quantitative implementation of this program requires a dedicated treatment. In particular, a dynamical c enters both BAO distance observables and CMB physics, including the sound horizon and recombination. We therefore leave a joint SNe–BAO–CMB analysis of the ζ - w CDM framework to future investigations.

8. Conclusion: Three physical implications of the dynamical- c cosmology

Figure 3 captures the central result of this Letter. In the w CDM plane, the post-ABC SNe and the H II-galaxy constraint of Chávez et al. (which is independent of YONSEI's ABC) both converge toward the Kolb point⁵. More broadly, the Kolb point belongs to a distinguished branch of the Dolgov–Barrow DSL family satisfying $\zeta = \frac{1}{2}(1 + 3w)$, the *scale-invariant geometry* (SIG) line. This embedding reveals a richer set of physical implications than those associated with the Kolb point alone. Figure 3 puts forth an alternative extension beyond w CDM where the speed of light, instead of dark energy, becomes dynamical.

Three implications of the SIG line in our ζ - w CDM are particularly noteworthy:

1. *The deceleration parameter is identical to the DSL index.* Since $q = (1 + 3w)/2$ for the corresponding single-fluid power-law cosmology, Eq. (14) immediately yields

$$q = \zeta \quad (18)$$

⁵ We note that the latter constitutes a *retrodiction*: the H II-galaxy data played no role in identifying the Kolb point in Nguyen (2026a).

along the SIG line. Hence $\zeta < 0, = 0, > 0$ corresponds respectively to accelerating, coasting, and decelerating power-law expansion. The expansion history is determined by the DSL index, $a(t) \propto t^{1/(1+\zeta)}$, solely specified by ζ . This is directly relevant to the ongoing debate (Son et al. 2025; Wiseman et al. 2026; Sah et al. 2026a,b; Ray et al. 2026).

2. *The comoving particle horizon is infinite.* Along the SIG line, $c/a \propto t^{-1}$, and hence

$$r_{\text{horizon}}(t) \propto \int_0^t \frac{dt'}{t'} \rightarrow \infty. \quad (19)$$

Thus the SIG cosmologies possess no finite comoving particle horizon, a property potentially relevant to early-time cosmology; in particular, the horizon paradox without invoking inflation (Moffat 1993; Albrecht & Magueijo 1999).

3. *The speed of light is proportional to the scale-factor growth rate.* Along the scale-invariant-geometry line, the Ricci scalar is strictly a^{-2} . Eq. (10) then yields the unique, nontrivial kinematic relation

$$\frac{aH}{c_{\text{dyn}}} = \text{const} \iff c_{\text{dyn}} = \frac{c_0}{a_0 H_0} \dot{a}, \quad (20)$$

a prediction that we recently advanced in Nguyen (2026b). Thus, along the SIG line toward which the post-ABC SNe and H II galaxies independently converge, *the dynamical speed of light and the scale-factor growth rate evolve in lockstep*.

Whether these recent empirical indications ultimately point to a dynamical-speed-of-light cosmology is an observational inquiry, and the proposed ζ - w CDM framework offers the BAO and CMB probes a concrete way to cast their vote.

Data Availability

The numerical python scripts used to generate the likelihood analyses and figures in Sects. 2 and 5 are publicly available at <https://github.com/HoangNguyenUBB/SNeIa-loga-relation>. The observational SNe Ia data used in this work are downloadable from the Pantheon+ and DES-SN5YR survey releases (Scolnic et al. 2022; Abbott et al. 2024).

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References

- Abbott, T. M. C., et al. (DES Collaboration) 2024, ApJ, 973, L14
- Albrecht, A. & Magueijo, J. 1999, Phys. Rev. D, 59, 043516
- Barrow, J. D. 1999, Phys. Rev. D, 59, 043515
- Chávez, R., Terlevich, R., González-Morán, A. L., et al. 2026, 2607.14254
- Chávez, R., Terlevich, R., Terlevich, E., et al. 2025, MNRAS, 538, 1264
- Chevallier, M., & Polarski, D. 2001, Int. J. Mod. Phys. D, 10, 213
- Chung, C., Son, J., Park, S., et al. 2026, MNRAS, 551, 1
- Copeland, E. J., Sami, M., & Tsujikawa, S. 2006, Int. J. Mod. Phys. D, 15, 1753
- Dolgov, A. D. 1997, Phys. Rev. D, 55, 5881
- Einstein, A. 1911, Annalen der Physik, 35, 898
- Fujii, Y. 1982, Phys. Rev. D 26, 2580
- González-Morán, A. L., Chávez, R., Terlevich, R., et al. 2019, MNRAS, 487, 4669
- Kolb, E. W. 1989, ApJ, 344, 543
- Linder, E. V. 2003, Phys. Rev. Lett., 90, 091301
- Magueijo, J. 2003, Rep. Prog. Phys., 66, 2025
- Mahtessian, A., Karapetian, G., Hovhannisyan, M., et al. 2026, A&A, 710, L8
- Mirpoorian, S., Lin, M.-X., & Pogosian, L. 2026, 2604.24761
- Moffat, J. W. 1993, Int. J. Mod. Phys. D, 2, 351

- Murakami, Y., Twedde, J., Wiseman, P., et al. 2026, 2604.16597
- Nguyen, H. K. 2025a, JCAP, 04, 005
- Nguyen, H. K. 2025b, EPJC, 85, 393
- Nguyen, H. K. 2025c, PLB, 862, 139357
- Nguyen, H. K. 2026a, 2608.00806
- Nguyen, H. K. 2026b, 2601.05512
- Perlmutter, S., Aldering, G., Goldhaber, G., et al. 1999, ApJ, 517, 565
- Popovic, B. & Sullivan, M. 2026, MNRAS, 550, 1
- Ray, S., Khlopov, M., Sanyal, A., et al. 2026, 2607.20570
- Riess, A. G., Filippenko, A., Challis, P., et al. 1998, AJ, 116, 1009
- Sah, A., Rameez, M., & Sarkar, S. 2026, MNRAS, 549, 1
- Sah, A., Rameez, M., & Sarkar, S. 2026, 2608.02484
- Scolnic, D., Brout, D., Carr, A., et al. 2022, ApJ, 938, 113
- Son, J., Lee, Y.-W., Chung, C., et al. 2025, MNRAS, 544, 975
- Terlevich, R., Terlevich, E., Melnick, J., et al. 2015, MNRAS, 451, 3001
- Wei, J.-J., & Melia, F. 2025, MNRAS, 542, L19
- Wetterich, C. 1988, Nucl. Phys. B 302, 668
- Wiseman, P., Popovic, B., Sullivan, M., et al. 2026, MNRAS, 549, 1
- Zhang, Y., Meng, X., Luo, J., et al. 2026, 2607.09199